Diseases of the Venous System

NURA 340 Pathophysiology | Lecture Study Guide

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Lecture roadmap: Venous physiology → Chronic Venous Insufficiency (CVI) → Deep Vein Thrombosis (DVT) → Pulmonary Embolism (PE) → Venous vs. Arterial comparison

The Veins: Normal Physiology

Having covered the high-pressure arterial system, we turn to the venous system. Veins are low-pressure, high-capacity vessels that hold roughly 60–70% of the body's total blood volume at any given time. Venous walls are thinner and contain far less smooth muscle than arterial walls, so blood return to the heart depends on three mechanisms:

- **One-way (bicuspid) valves:** folds of the tunica intima that open to allow forward flow toward the heart and snap shut to prevent retrograde (backward) flow under gravity.
- **Skeletal muscle pump:** contraction of the deep leg muscles (e.g., the calf) compresses the veins and drives blood upward against gravity.
- **Respiratory pump:** diaphragm contraction during inhalation decreases intrathoracic pressure and increases intra-abdominal pressure, pulling blood up into the vena cava.

Key concept: Venous disease occurs when these forward-flow mechanisms fail. The result is blood pooling, increased venous wall tension, vessel wall inflammation, and thrombus formation. Keep these three mechanisms in mind for every condition below.

Chronic Venous Insufficiency (CVI)

When damage to deep or superficial veins persists over months to years, the patient develops CVI: a sustained structural disorder in which the veins cannot adequately return blood to the heart. CVI is the venous counterpart of peripheral arterial disease (PAD), which we covered previously.

Pathophysiology

The core driver is **venous hypertension in the lower extremities**.

1. **Valve failure:** Stretched venous walls or damage from a previous DVT prevent the valve leaflets from meeting, so blood flows backward under gravity.
2. **Pressure transfer:** Sustained high pressure is transmitted downward into the distal microvessels and skin capillaries.
3. **Fluid shift:** Elevated capillary hydrostatic pressure pushes fluid, plasma proteins, and red blood cells into the dermis and subcutaneous tissue.
4. **Hyperpigmentation:** Extravasated red blood cells break down and release iron pigment (hemosiderin), turning the lower leg skin a characteristic dark brown.
5. **Fibrosis:** Chronic inflammation stimulates dermal fibrosis, so the skin and subcutaneous tissue become firm, "woody," and constricted.

6. **Venous stasis and ischemia:** Local tissue edema impairs capillary oxygen diffusion, leading to local cutaneous ischemia, cell death, and ulcer breakdown.
7. **Lymphatic overload:** The elevated capillary hydrostatic pressure overwhelms interstitial lymphatic drainage, producing chronic low-grade lymphedema.
8. **Barrier breakdown:** Fluid accumulation physically separates skin cells from the microvascular nutrient beds, degrading skin barrier function and perpetuating inflammation.
9. **Outcome:** Tissue necrosis and ulceration.

Risk Factors

- Previous DVT
- Chronic standing or sitting (occupations with prolonged standing or prolonged sitting)
- Advanced age
- Gender: females are at higher risk than males
- Obesity
- Family history (especially of varicose veins, which are superficial veins)

Progression: Varicose Veins to Venous Stasis Ulcers

High pressure overflowing through incompetent perforator veins into the superficial system produces dilated superficial veins (varicose veins). Over time, this progresses through leg edema and skin changes to venous stasis ulcers.

Venous stasis ulcer characteristics: painless or mildly tender; shallow with irregular borders; moist, ruddy red granulation base; heavy exudate; typically located at the **medial malleolus**.

Signs and Symptoms

- **Bilateral lower extremity edema:** typically affects both legs (not one over the other) and the lower extremities far more than the upper. Early on it is soft and pitting; over time it becomes firm, non-pitting, "brawny" edema.
- **Dull, aching, heavy sensation:** legs feel fatigued or throbbing; worse with prolonged standing and improved by elevating the legs above heart level (elevation aids return against gravity).
- **Stasis dermatitis:** pruritic, dry, scaly, reddened skin on the lower legs.
- **Hemosiderin staining:** dark brown, bronze, or hyperpigmented discoloration around the ankles and shins.

Don't confuse with PAD: The dark discoloration of CVI comes from fluid and blood pooling due to poor venous return, not from low perfusion. Venous ulcers are red and weepy (fluid shifts); arterial ulcers are dry, pale, or black (no perfusion). In CVI there is blood in the leg, but it is not returning to the heart.

Diagnostic Tests

- **Ultrasound:** the gold standard imaging test. It evaluates valve closure by measuring the duration of retrograde blood flow.
- **Air plethysmography:** measures leg volume changes during positional shifts (sitting vs. standing) to assess overall venous pump efficiency.

Complications

Decreased quality of life, ranging from mild discomfort and heaviness to open wounds. This carries a higher risk of depressive disorders and anxiety.

Deep Vein Thrombosis (DVT)

DVT is the formation of a blood clot within a deep vein, most commonly in the large vessels of the lower extremities: the **popliteal, femoral, or iliac veins**. (In practice, clinicians almost always just say "DVT.")

Virchow's Triad

A DVT almost never occurs spontaneously without at least two of these three factors. Memorize this triad.

Factor	What happens
Venous stasis	Slow or stagnant flow lets platelets and clotting factors accumulate in a quiet valve pocket of the deep veins, where local hypoxia develops.
Endothelial injury	Direct damage to the vessel intima (innermost layer) exposes subendothelial collagen and tissue factor, triggering the intrinsic coagulation cascade and attracting circulating platelets.
Hypercoagulability	A systemic shift toward prothrombotic factors: increased fibrinogen, enhanced thrombin production, or diminished natural anticoagulants (protein C, protein S, antithrombin III).

Pathophysiology

Once the process is triggered, a fibrin mesh forms and traps red blood cells and platelets. The clot grows in the direction of blood flow, partially or completely occluding the venous lumen and provoking a localized inflammatory response in the vessel wall (thrombophlebitis).

DVT is an inflammatory-thrombotic process: localized venous stasis causes endothelial cell hypoxia, which upregulates adhesion molecules and attracts leukocytes. Inflammatory cytokines then stimulate tissue factor release, directly linking systemic inflammation to activation of the clotting cascade.

Risk Factors

- **Prolonged immobility:** bed rest, long flights or car rides, post-operative states, and long time in the OR.
- **Active malignancy**
- **Pregnancy and postpartum periods**
- **Oral contraceptives or estrogen replacement therapy:** a major risk factor. Combined with pregnancy/postpartum risk, DVTs are seen more often in female than male patients.
- **Obesity and advanced age**
- **Inherited thrombophilias:** you don't need to know specific ones, only that they create a hypercoagulable state.

Nursing connection: Sequential compression devices (SCDs) are placed on immobile hospitalized patients to compress the legs and prevent blood pooling and venous stasis, which removes one leg of Virchow's triad.

Signs and Symptoms

- **Unilateral presentation:** usually only one leg is affected, since a clot forming in both legs at once is uncommon.
- **Edema and swelling distal to the occlusion:** the obstructed vein blocks return to the heart, blood pools below the clot, venous pressure rises, and the increased hydrostatic pressure forces fluid into the interstitial space.
- **Localized pain and tenderness:** dull ache or tightness in the calf or thigh, worse with weight bearing or palpation over the deep vein tract.
- **Erythema and warmth:** redness and localized heat from inflammatory vasodilation of surrounding cutaneous tissue.
- **Superficial vein dilation:** collateral veins around the blocked vessel dilate to bypass the occlusion and carry blood back toward the heart.

Diagnostic Tests

- **Ultrasound (first line):** shows Doppler blood flow and the venous segment, and can sometimes show the clot itself blocking flow.
- **D-dimer:** detects cross-linked fibrin produced when plasmin breaks down a clot. A negative D-dimer safely rules out DVT in low-probability patients. A positive result is nonspecific (it can rise with surgery, cancer, pregnancy, trauma, or inflammation), but it warrants further investigation for DVT.

Complications

- Prolonged hospitalization, which can bring immobility, deconditioning, and secondary infections.
- **Post-thrombotic syndrome:** a long-term disorder with manifestations similar to CVI. This is why DVT is a risk factor for CVI.
- **Pulmonary embolism:** the acute, life-threatening complication (next section).

Pulmonary Embolism (PE)

A PE is an acute, life-threatening consequence of conditions like DVT. A fragment of a thrombus (or an entire clot) detaches from the deep vein wall, travels through the venous network to the right atrium, then the right ventricle, and is pumped into the pulmonary arterial tree, where it lodges and blocks blood flow.

Pathophysiologic Consequences

PE affects both the respiratory and hemodynamic systems.

- **V/Q mismatch:** the blocked pulmonary artery stops blood flow to an area of alveolar tissue. The alveoli are still ventilated, but with no blood flow there is no gas exchange, creating alveolar dead space. The location of the clot determines which area is affected.

- **Bronchoconstriction and hypoxemia:** local ischemia triggers release of histamine, serotonin, and inflammatory prostaglandins from platelets, causing reflex bronchoconstriction and atelectasis and driving acute arterial hypoxemia.
- **Increased pulmonary vascular resistance:** the physical blockage plus vasoconstriction increases the workload of the right side of the heart.
- **Right ventricular failure:** the thin-walled RV is designed for low pressures. When forced to pump against massive pulmonary pressure it dilates, and the interventricular septum bulges toward the left ventricle. This impairs LV filling, drops stroke volume, and leads to systemic hypotension and obstructive shock.

Acid-Base Changes

- **Early:** acute hypoxia and irritant receptor stimulation cause hyperventilation, so the patient blows off CO_2 and develops respiratory alkalosis.
- **Late:** as obstructive shock worsens, systemic perfusion fails and anaerobic metabolism triggers a secondary metabolic acidosis.

Know this: A pulmonary embolism can cause respiratory alkalosis.

Risk Factors / Causes

- **DVT:** risk factors are identical to DVT (Virchow's triad). About 90% of PEs originate from deep vein thrombi.
- **Air bubbles:** air introduced during IV fluid or medication administration. This is why you will learn to prime IV tubing and remove all air from the line.

Signs and Symptoms

Unlike a DVT, which can develop gradually, PE onset is typically sudden.

- **Sudden-onset dyspnea:** the single most common symptom. It can occur instantly, even at rest, and is unrelated to activity level.
- **Tachypnea and tachycardia:** compensatory sympathetic responses to hypoxia and falling stroke volume.
- **Pleuritic chest pain:** sharp, knife-like pain on inspiration, caused by localized pulmonary infarction and pleural inflammation.
- **Sense of impending doom:** apprehension that something is very wrong, from acute systemic hypoxemia and sympathetic surge. Symptoms often snowball: dyspnea, then tachycardia and tachypnea, then anxiety and a feeling of doom as the sympathetic surge builds.
- **Hemoptysis:** coughing up blood-tinged sputum, indicating alveolar hemorrhage secondary to pulmonary infarction.
- **Signs of RV failure:** jugular vein distension, hypotension, syncope, altered mental status, and potentially cardiac arrest. PE can kill the patient, which is why it is an emergency.

Diagnostic Tests

- **CT pulmonary angiography (CTPA):** the gold standard. Contrast fills the pulmonary vascular tree and clearly shows the location of the occlusion.
- **Ventilation/perfusion (V/Q) scan:** a nuclear medicine alternative if needed.

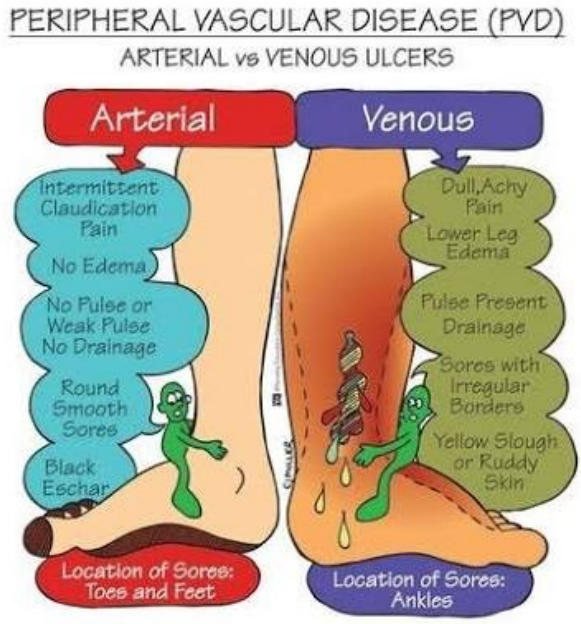
Labs (drawn because the cause is not always clear in the moment, so other possibilities must be ruled out):

- **Arterial blood gas (ABG):** acute arterial hypoxemia (PaO_2 < 80 mmHg) and acute hypocapnia (low CO_2), indicating early respiratory alkalosis.
- **D-dimer:** a negative result safely excludes PE risk.
- **Troponin:** elevation indicates right ventricular microvascular ischemia (heart wall strain from lack of blood flow).
- **BNP (brain natriuretic peptide):** elevation confirms acute ventricular stretch and strain. BNP can reflect left or right strain, but in the setting of PE we know it is right-sided.
- **EKG:** patients present with chest pain, dyspnea, tachycardia, and tachypnea, so the EKG helps rule out other causes.

Venous vs. Arterial Disease

Contrast how arterial and venous disease mechanisms present physically at the bedside, particularly peripheral arterial disease versus chronic venous insufficiency. Being able to differentiate them helps you identify what is happening quickly, prioritize nursing interventions, know what is appropriate in the care of your client, and know what to include in patient education. Understanding the structural differences lets you link underlying vascular physiology to assessment findings and diagnostic workups.

	Arterial	Venous
Pain	Intermittent claudication	Dull, achy pain
Edema	No edema	Lower leg edema
Pulses	No pulse or weak pulse	Pulse present
Drainage	No drainage	Drainage (weepy, heavy exudate)
Ulcer appearance	Round, smooth sores; black eschar	Irregular borders; yellow slough or ruddy skin
Ulcer location	Toes and feet	Ankles (medial malleolus)



Source: lecture slide 9

Quick Review: Side-by-Side

	CVI	DVT	PE
Definition	Sustained failure of veins to return blood to the heart	Clot in a deep vein (popliteal, femoral, iliac)	Clot lodges in the pulmonary arterial tree
Key mechanism	Venous hypertension from incompetent valves	Virchow's triad	Embolus from DVT (about 90%) or air bubble
Presentation	Bilateral edema, aching, hemosiderin staining, stasis dermatitis, ulcers	Unilateral swelling, pain, warmth, redness	Sudden dyspnea, tachypnea/tachycardia, pleuritic pain, doom, hemoptysis
Diagnostics	Ultrasound (gold standard), air plethysmography	Ultrasound (first line), D-dimer	CTPA (gold standard), V/Q scan, ABG, D-dimer, troponin, BNP, EKG

Notes on Transcript Cleanup

The following terms were mis-transcribed and were corrected using lecture context and the slides: "epithelial injury" is endothelial injury; "hemosiderian" is hemosiderin; "medial moellus" is medial malleolus; "lymphedemia" is lymphedema; "air plesmiography" is air plethysmography; "thrombolytis" is thrombophlebitis. Please verify the PaO_2 threshold, and that the professor wants "incompetent perforator veins," against your slides or notes.